SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I S.K. Deepa Arathi 192571020 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

S.K. Deepa

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

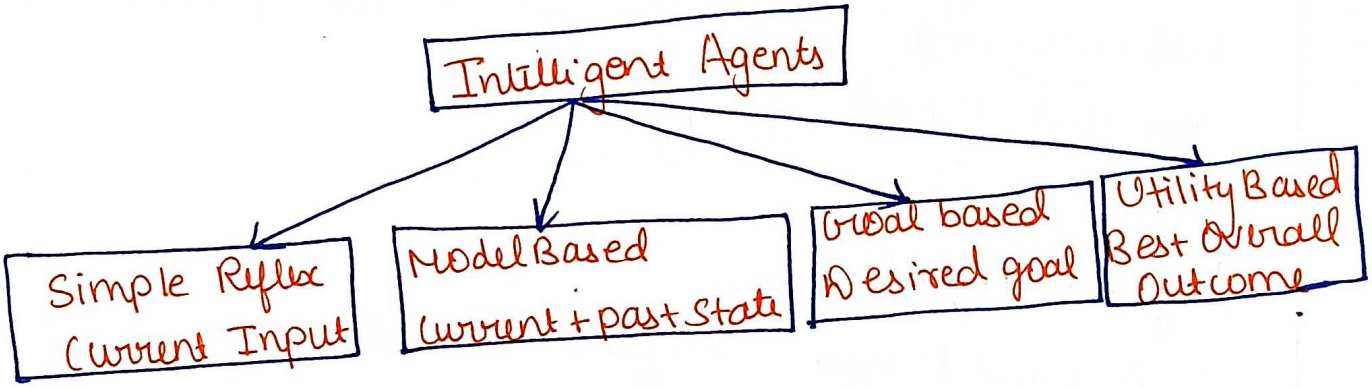
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.
   Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Introduction:-

A smart home automation System is an AI-based system that observes its environment through sensors and control devices through actuators. Inputs may include motion, temperature, time of day, door/window status and security events



Simple Reflex Agent:-

A Simple Reflex agent makes decisions based only on the current input using condition action rules. It does not store past information. For example it switches on a light when motion is detected.

Moded Based Agent:-

A model based agent maintains an internal state to understand the environment using current inputs and past information. For example, a smart home system checks room occupancy and window status before controlling the air conditioner.

Goal Based Agent:-

A goal based agent selects actions to achieve a specific goal. It considers different possible actions and chooses one that helps reach desire result.

Utility Based Agents:-

A Utility Based agent evaluates different possible outcomes and chooses the action that provides the greatest overall benefits.

Hybrid Agent:-

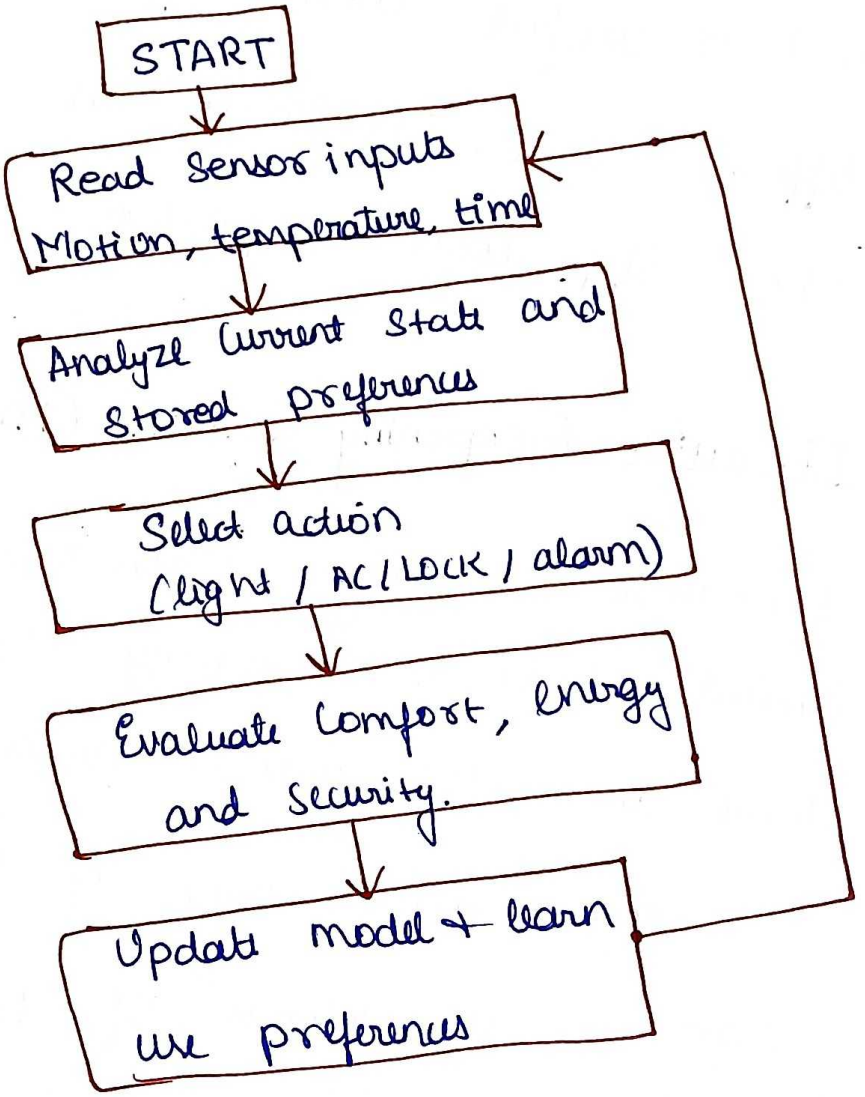
A hybrid agent combines two or more types of agents to make better decisions. For example, a Smart home system can respond quickly to emergencies.

Comparison:-

Agent	Decision basis	Smart home example	Limitation
Simple Reflex	Current input	Motion → Light ON	No memory
Model based	Current + stored state	Check Occupency	More complex
Goal Based	Desired goal	Maintain temperature	Copeting goal
Utility Based	Best Overall Outcome	Balance Comfort + energy	Utility design

Recommended approach :-

A hybrid agent is most effective approach becaus it combins different agent types. It provids quick responses remembers, past information, achievs goal and balancs comfort, safety and energy efficiency.



Conclusion:-

Intelligent agents help smart home systems make efficient decisions, improve safety and enhanes comfort. A hybrid agent provids bettter performance by combining Stoength of different agent types.

2)

Uniform Cost Search (UCS)

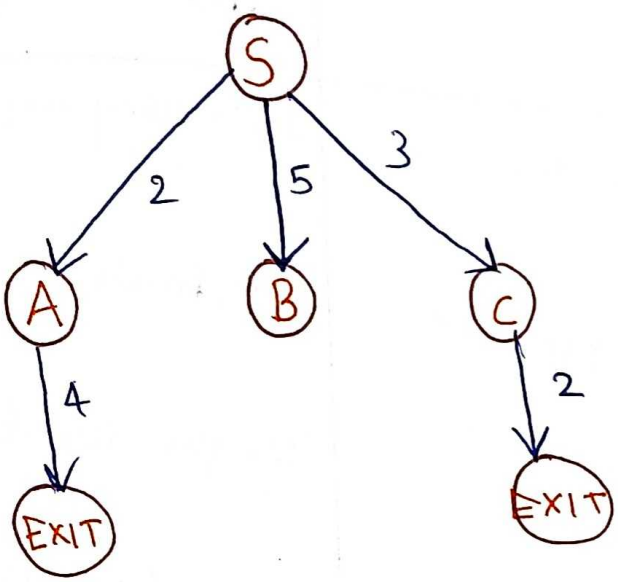
Uniform cost search expands the node with the lowest accumulated path cost first. It uses a priority queue to select the least cost path and is useful when different action have different cost. It finds an optimal solution when step costs are positive.

Iterative Deepening Search (IDS)

Iterative Deepening Search repeatedly performs depth limited searches, increasing the depth limit each time until the goal is found. It combines the low memory usage of Depth first search with the completeness of Breadth first search.

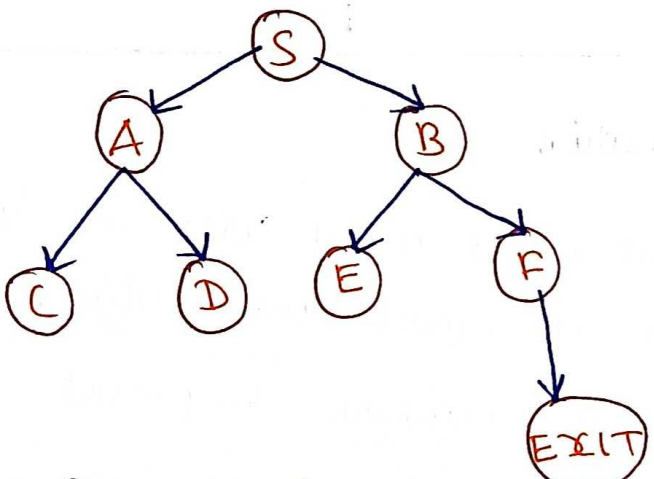
It is useful when the solution depth is unknown and memory is limited.

UCS Example



Least cost $S \rightarrow C \rightarrow$ Exit $= 5$

IDS Example



Depth Limits $0 \rightarrow 1 \rightarrow 2 \rightarrow 3$ (exit found)

UCS (VS) IDS

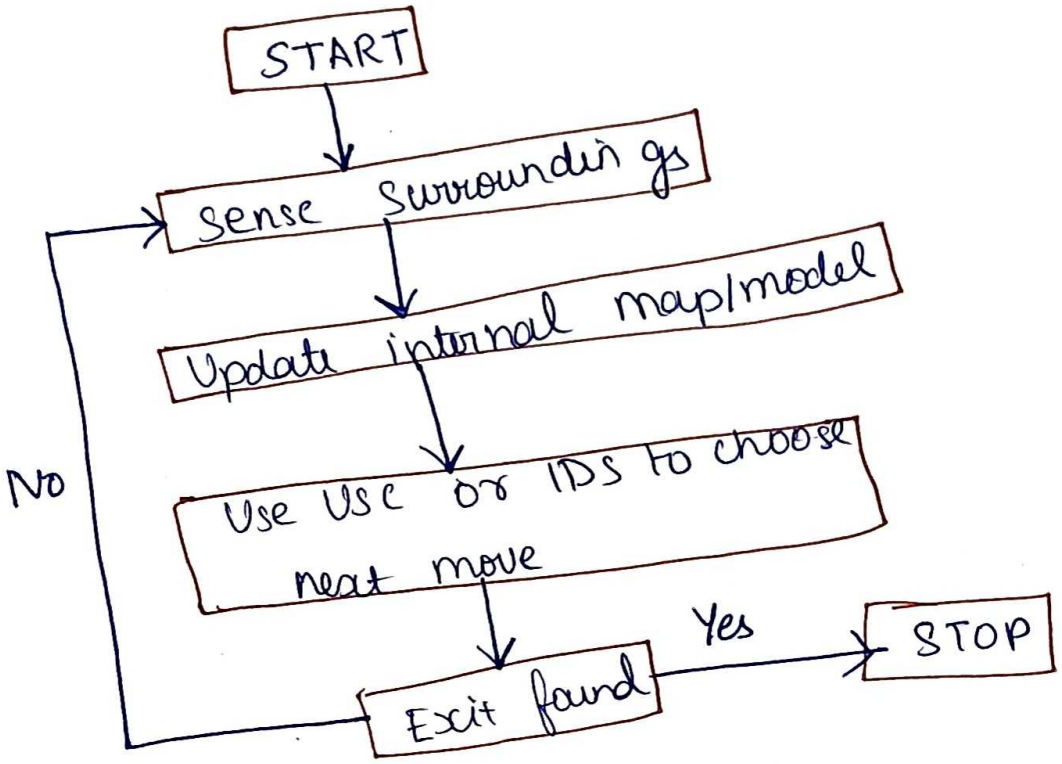
Feature	UCS	IDS
Expansion rule	Lower path cost first	Increasing depth limit
Complete	Yes with positive cost	Yes, finite branching
Optimal	Yes	Yes for equal step cost
Time	can be high, cost dependent	O(b^d)
Space	High: cost dependent	O(bd)
Best Use	Different movement cost	Equal cost

Most Effective Combination

A model based + goal based agent with UCS is generally the best choice when movement costs differ. The model based component remembers explored areas, the goal based components focuses on reaching the exit and UCS find least cost route.

Relation to Intelligent Agents

* Simple reflex
* model based
* Goal based



Conclusion:-

Agent design and search Strategy should work together in an unknown rate, memory of explored areas prevents repeated work, the goal directs the robot towards the exit. and UCS & IDS provides Systematic Search method.